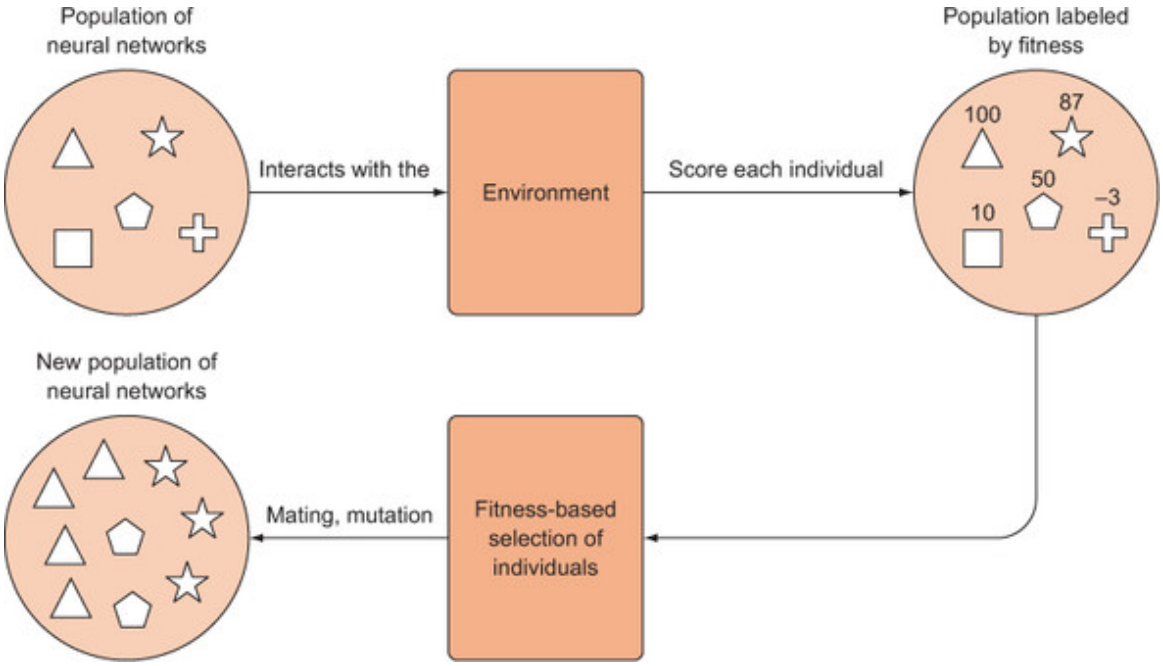


4. The individuals in the breeding population will then “mate” to produce “offspring” that will form a new, full population of 100 individuals. If the individuals are simply parameter vectors of real numbers, mating vector 1 with vector 2 involves taking a subset from vector 1 and combining it with a complementary subset of vector 2 to make a new offspring vector of the same dimensions. For example, suppose you have vector 1: [1 2 3] and vector 2: [4 5 6]. Vector 1 mates with vector 2 to produce [1 5 6] and [4 2 3]. We simply randomly pair up individuals from the breeding populations and recombine them to produce two new offspring until we fill up a new population. This creates new “genetic” diversity with the best performers.
5. We now have a new population with the top solutions from the last generation, along with new offspring solutions. At this point, we will iterate over our solutions and randomly mutate some of them to make sure we introduce new genetic diversity into every generation to prevent premature convergence on a local optimum. Mutation simply means adding a little random noise to the parameter vectors. If these were binary vectors, mutation would mean randomly flipping a few bits; otherwise we might add some Gaussian noise. The mutation rate needs to be fairly low, or we’ll risk ruining the already present good solutions.
6. We now have a new population of mutated offspring from the previous generation. We repeat this process with the new population for *N*